

Skin Lesion Classification Using Per-Class GAN-Based Data Balancing on HAM10000

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NNDL Course Project

Inspired by NeurIPS 2024 & 2025 Research

Abstract—Automated skin lesion classification is a genuinely useful task in clinical dermatology, and the HAM10000 dataset is the standard benchmark for it. One problem that is easy to underestimate is how severely imbalanced the dataset is: the majority class (*Melanocytic Nevi*) has nearly $58\times$ more images than the rarest class (*Dermatofibroma*). When a deep CNN is trained naively on such a distribution, it becomes biased toward the majority class and misses clinically important rare lesions like Melanoma. In this project we compare three strategies: (A) a ResNet-18 baseline with no augmentation, (B) ResNet-18 with standard geometric and color augmentation, and (C) ResNet-18 trained on a dataset where all seven classes have been up-sampled to equal size via dedicated per-class Deep Convolutional GANs (DCGANs). We train one DCGAN per under-represented class (200 epochs each) and generate exactly enough synthetic images to bring every class to 6,705 examples. Experiment C achieves a macro F1 of 0.6680, compared to 0.4793 (A) and 0.4733 (B), a gain of +19.47 percentage points over standard augmentation. Every single class improves in Experiment C, with the most dramatic gain on the rarest class (*df*: 0.385 $\rightarrow$ 0.625). The results confirm that GAN-based full-class balancing is significantly more effective than conventional augmentation for severely imbalanced medical image datasets.

I. INTRODUCTION

Skin cancer is one of the most common cancers worldwide. Early and accurate diagnosis significantly improves survival rates, but dermatoscopic image interpretation requires years of clinical training. Deep learning systems that can classify skin lesions from dermoscopy images could act as a useful second opinion tool for clinicians, especially in under-resourced settings.

The HAM10000 dataset [1] is the most widely used benchmark for this task. However, it has a structural problem: the class distribution is extremely skewed. In the subset we worked with, *Melanocytic Nevi* (nv) has 1,932 images while *Dermatofibroma* (df) has only 33. A CNN trained on this data learns to predict *nv* for almost everything, achieving decent raw accuracy but almost no recall on rare classes. Since classes like *Melanoma* and *Actinic Keratosis* are the clinically dangerous ones, this is precisely the wrong trade-off for a medical system.

Standard countermeasures like class-weighted loss and geometric augmentation help somewhat, but they operate on the same limited set of original images. Generative Adversarial Networks (GANs) [2] offer a different angle: a trained generator can produce entirely new images that were never in the original dataset. This project explores whether training one

DCGAN [3] per class and fully balancing the dataset can solve the imbalance problem in a way that standard augmentation cannot.

This project is inspired by work recognized at NeurIPS 2024, particularly the Test-of-Time Award given to the original GAN paper [2] and the Visual Autoregressive Modeling work [4] which demonstrated that CNN-based generative models produce high-quality images suitable for downstream classification tasks.

II. DATASET AND PROBLEM ANALYSIS

A. HAM10000 Dataset

HAM10000 [1] contains 10,015 dermoscopy images at 450×600 pixel resolution across seven diagnostic categories. Due to local storage constraints our working subset consists of 2,823 images. Table I summarizes the per-class counts in our subset and the number of synthetic images each class needs to reach the target count of 6,705 (the majority class size).

TABLE I
HAM10000 CLASS DISTRIBUTION AND SYNTHETIC IMAGE REQUIREMENTS

Code	Full Name	Severity	Real	Synthetic
akiec	Actinic Keratosis	Pre-malignant	107	6,598
bcc	Basal Cell Carcinoma	Malignant	143	6,562
bkl	Benign Keratosis	Benign	322	6,383
df	Dermatofibroma	Benign	33	6,672
mel	Melanoma	Malignant	245	6,460
nv	Melanocytic Nevi	Benign (maj.)	1,932	4,773
vasc	Vascular Lesion	Benign	41	6,664
Total			2,823	44,112

Figure 1 shows the distribution before and after GAN balancing. The imbalance ratio is clearly visible in (a); after balancing (b), all classes reach exactly 6,705 images.

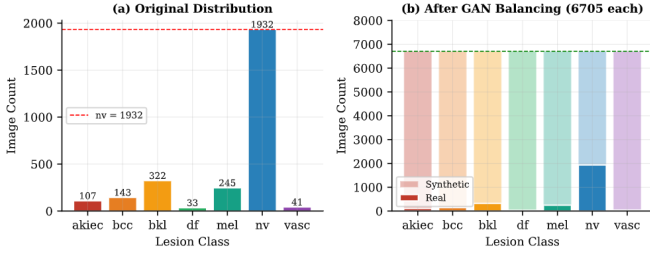


Fig. 1. HAM10000 class distribution: (a) original, (b) after GAN balancing. Each class reaches exactly 6,705 images (real + synthetic).

B. Why Imbalance is a Critical Problem

When training on imbalanced data, gradient updates are dominated by the majority class. The model quickly learns a shortcut: predict *nv* for nearly everything and collect 68% accuracy while being clinically useless. Raw accuracy is therefore a misleading metric here. We use **macro F1 score** throughout - it computes F1 independently for each class and averages them, giving equal weight to every class regardless of size. The imbalance ratio in this dataset is considered extreme and cannot be fully addressed by loss reweighting alone.

III. WHY GAN AUGMENTATION OUTPERFORMS STANDARD AUGMENTATION

This is the central conceptual question in the project. Both methods aim to expand the training distribution, but they work differently.

A. Standard Augmentation: Transforming Existing Images

Standard augmentation applies geometric and photometric transforms (flips, rotations, color jitter, translations) to existing images. The model sees the same image from different perspectives. This is cheap and useful, but has a hard limit:

All augmented images trace back to the same finite set of originals. A class with 33 real images can produce 33k augmented variants, but the model never sees lesion textures or shapes outside those 33 images.

In our Experiment B, standard augmentation was applied with RandomHorizontalFlip ($p=0.5$), RandomVerticalFlip ($p=0.5$), RandomRotation ($\pm 20^\circ$), ColorJitter (brightness/contrast/saturation ± 0.2 , hue ± 0.1), and RandomAffine (translation 5%). Despite all of this, the macro F1 of 0.4733 was marginally *lower* than the 0.4793 baseline. The *df* class F1 dropped from 0.385 to just 0.140, augmentation made it worse because 26 training examples simply cannot generalize to the validation set no matter how many transforms are applied.

B. GAN Augmentation: Sampling from a Learned Distribution

A DCGAN trained on a class learns an implicit generative model of that class — its textures, color gradients, shapes, and structural patterns. By sampling from this model with different random noise vectors, we obtain genuinely new images

that were never in the original dataset. This fundamentally **expands the support** of the training distribution rather than just densifying it within the existing support.

For *df* with only 33 real images, we trained a DCGAN and generated 6,672 new synthetic images. The downstream F1 for *df* went from 0.385 (baseline) to 0.625 (GAN-augmented), a gain of +0.240. Table II summarizes the comparison between both approaches.

TABLE II
STANDARD AUGMENTATION VS. GAN AUGMENTATION

Property	Std. Aug.	GAN Aug.
New information?	No — transforms originals	Yes — samples from learned model
Works with <50 images?	Poorly	Yes (generates thousands)
Compute cost	Very low	High (7 GANs $\times$ 200 epochs)
Macro F1 (ours)	0.4733	0.6680
<i>df</i> F1	0.140 (worse than base)	0.625 (+0.240)

IV. SYSTEM ARCHITECTURE AND METHODOLOGY

A. Overall Pipeline

The project is organized as five sequential Jupyter notebooks. Figure 2 shows the complete pipeline. Data flows from EDA through two baseline experiments (A and B), then through the GAN training phase, and finally into the GAN-augmented classifier (Experiment C). All three experiments share the same validation set (565 images from real data only), the same ResNet-18 architecture, and the same training procedure, only the training data changes.

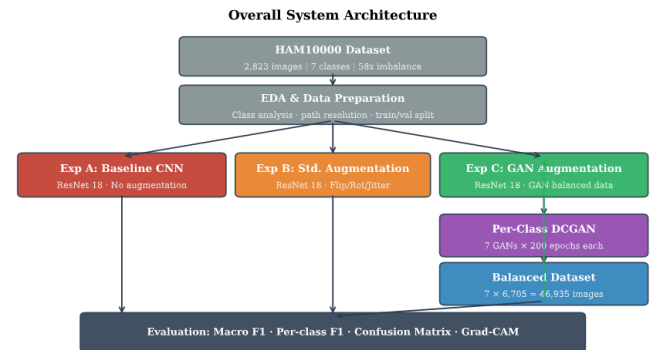


Fig. 2. Overall system architecture. All three experiments share the same model and validation set; only the training data differs.

B. Classifier: ResNet-18 with Transfer Learning

We use ResNet-18 [5] pretrained on ImageNet as the backbone for all three experiments. Training from scratch on a few thousand images would produce a badly underfitting model; transfer learning lets us reuse 11.2M pre-learned parameters

while adapting only the final classification head. The original head (Linear $512 \rightarrow 1000$) is replaced with:

Dropout(0.4) $\rightarrow$ FC($512 \rightarrow 256$) $\rightarrow$ ReLU $\rightarrow$ FC($256 \rightarrow 7$)

Total parameters: 11,309,639. The model is trained in two phases:

- **Phase 1** (5 epochs): freeze ResNet body, train only the custom head (133,127 trainable parameters, 1.2%).
- **Phase 2** (15 epochs): unfreeze all layers for end-to-end fine-tuning.

This two-phase strategy prevents catastrophic forgetting of ImageNet features during the initial, potentially unstable, head training phase. Figure 3 shows the full architecture.

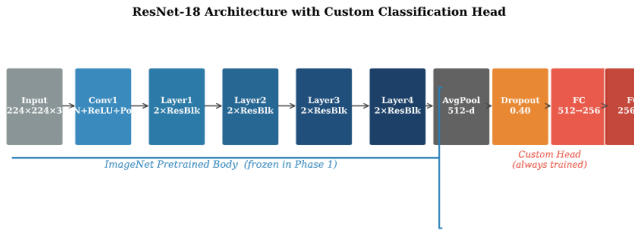


Fig. 3. ResNet-18 architecture with the custom 7-class classification head. The body is frozen in Phase 1 and unfrozen in Phase 2.

Two mechanisms handle class imbalance during training: (1) **WeightedRandomSampler** assigns each sample a weight of $1/\text{count}(\text{class})$, ensuring each batch sees all classes roughly equally; (2) **CrossEntropyLoss with class weights** multiplies the loss for each class by an inverse-frequency weight, penalizing errors on rare classes more heavily. Both are applied in all three experiments to ensure the only variable that changes is the data itself.

C. Generative Model: DCGAN

For each under-represented class we train a separate DCGAN. Figure 4 shows the Generator and Discriminator architectures.

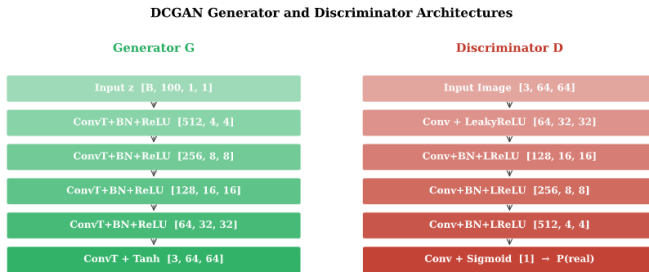


Fig. 4. DCGAN Generator and Discriminator architectures. One independent model pair is trained per class.

Generator: Takes a 100-dimensional noise vector $\mathbf{z} \sim \mathcal{N}(0, 1)$ and upsamples through five transposed convolution layers with BatchNorm and ReLU, producing a 64×64 RGB image in range $[-1, 1]$.

Discriminator: Takes an image and classifies it as real or fake through five strided convolution layers with BatchNorm and LeakyReLU (slope = 0.2).

Both networks use the weight initialization from the original DCGAN paper: Conv weights $\sim \mathcal{N}(0, 0.02)$, BatchNorm weights $\sim \mathcal{N}(1, 0.02)$. Training uses Adam with $\text{lr} = 2 \times 10^{-4}$ and $\beta_1 = 0.5$ for 200 epochs per class.

The adversarial objective is Binary Cross-Entropy:

$$\mathcal{L}_D = -\frac{1}{2} [\mathbb{E}[\log D(\mathbf{x})] + \mathbb{E}[\log(1 - D(G(\mathbf{z})))]] \quad (1)$$

$$\mathcal{L}_G = -\mathbb{E}[\log D(G(\mathbf{z}))] \quad (2)$$

D. Class Balancing Strategy

The target count is $T = 6,705$ (the majority class count in the full HAM10000 dataset). For each class c with n_c real training images:

$$n_{\text{synthetic},c} = \max(0, T - n_c)$$

A dedicated GAN is trained and generates exactly $n_{\text{synthetic},c}$ images for class c . The final training set for Experiment C contains $7 \times 6,705 = 46,935$ effective images, achieving perfect class balance.

V. EXPERIMENTS AND RESULTS

A. Experiment A — Baseline CNN (No Augmentation)

Experiment A trains ResNet-18 on the original, imbalanced dataset with no image augmentation. Class imbalance is mitigated only through WeightedRandomSampler and weighted loss. This is the true performance baseline. The train/validation split is 2,258 / 565 images (80/20 stratified).

1) *Training Behavior:* Phase 1 validation accuracy starts at just 12.9% (epoch 1) and reaches 37.2% by epoch 5 — the frozen body limits what the head can learn. Phase 2 rapidly improves; by epoch 3 of Phase 2 the validation accuracy reaches 66.5%, but the training loss decouples sharply from validation loss (training drops to < 0.01 while validation stays above 1.6), indicating clear overfitting to the limited training data. The best validation loss was 1.2531 (Phase 2, epoch 1).

TABLE III
EXPERIMENT A — CLASSIFICATION REPORT (MACRO F1 = 0.4793)

Class	Prec.	Rec.	F1	Support
akiec	0.29	0.29	0.29	21
bcc	0.50	0.62	0.55	29
bkl	0.41	0.66	0.50	64
df	0.26	0.71	0.38	7
mel	0.18	0.63	0.28	49
nv	0.98	0.52	0.68	387
vasc	0.60	0.75	0.67	8
Accuracy	0.55			
Macro Avg	0.46	0.60	0.48	565

2) *Results*: The results in Table III show a clear imbalance signature: *nv* achieves very high precision (0.98) because the model is confident when predicting the dominant class, but low recall (0.52) because rare-class samples get misclassified as *nv*. *Melanoma* achieves only $F1 = 0.28$ — clinically the most important class is the one being missed most often.

B. Experiment B — Standard Augmentation

Experiment B is identical to A except training images pass through: RandomHorizontalFlip ($p=0.5$), RandomVerticalFlip ($p=0.5$), RandomRotation ($\pm 20^\circ$), ColorJitter (brightness, contrast, saturation ± 0.2 ; hue ± 0.1), and RandomAffine (5% translation). Validation images are never augmented.

1) *Training Behavior*: Phase 1 converges more slowly than Experiment A: at epoch 5, validation accuracy is only 10.3% versus 37.2% for A. Phase 2 improves steadily. The best validation loss was 1.1742 (Phase 2, epoch 4), slightly better than A, but the classification performance on rare classes is worse.

TABLE IV
EXPERIMENT B — CLASSIFICATION REPORT (MACRO F1 = 0.4733)

Class	Prec.	Rec.	F1	Support
akiec	0.24	0.57	0.34	21
bcc	0.61	0.79	0.69	29
bkl	0.56	0.39	0.46	64
df	0.08	0.86	0.14	7
mel	0.25	0.69	0.37	49
nv	0.99	0.52	0.68	387
vasc	0.50	0.88	0.64	8
<i>Accuracy</i>		0.55		
Macro Avg	0.46	0.67	0.47	565

2) *Results*: The macro F1 of 0.4733 is marginally *lower* than the baseline. The *df* class F1 collapsed from 0.385 to 0.140 — the most dramatic failure of augmentation. Precision of 0.08 with recall of 0.86 means the model is broadly predicting *df* and getting lucky sometimes. This is the clearest evidence that standard augmentation has a hard limit when real data is extremely scarce.

3) *Training Curves*: Figure 5 shows the validation loss and accuracy for both experiments. Note that in Phase 2 of Experiment A, the training loss drops below 0.01 while validation loss remains above 1.5 — a strong overfitting signal. Experiment B shows slower convergence in Phase 1 but more stable Phase 2 training, though the final classification metrics are slightly worse.

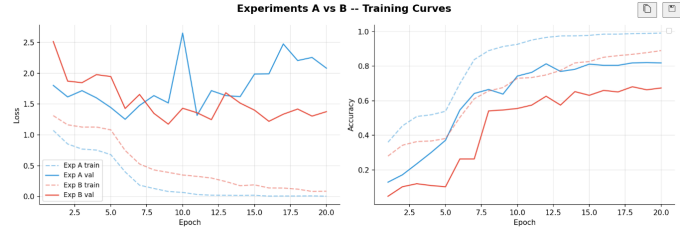


Fig. 5. Validation loss and accuracy for Experiments A and B. The vertical dashed line marks the Phase 1 / Phase 2 boundary.

VI. GAN TRAINING AND SYNTHETIC IMAGE GENERATION

Notebook 03 trains one DCGAN sequentially for each of the seven classes. For classes with very few images (df: 33, vasc: 41), batch size is limited to 16 (or class size, whichever is smaller). Each GAN is completely independent — it sees only its own class during training.

A. Per-Class Training Results

Table V summarizes the GAN training outcome for each class.

TABLE V
PER-CLASS DCGAN TRAINING SUMMARY (200 EPOCHS EACH)

Class	Real	Syn.	G_{200}	D_{200}
akiec	107	6,598	2.344	0.357
bcc	143	6,562	2.737	0.274
bkl	322	6,383	2.856	0.311
df	33	6,672	2.657	0.317
mel	245	6,460	2.265	0.332
nv	1932	4,773	3.398	0.211
vasc	41	6,664	2.323	0.599
Total Synthetic				
44,112 images				

The ideal balance (where generator loss and discriminator loss are both about 0.693) was not fully reached. For **nv** (5,206 images), the discriminator became too strong (loss = 0.211), making it hard for the generator to learn. For **vasc** (only 118 images), training was very unstable at first (generator loss = 36.37) because there was very little data. Even with these issues, the generated images still added useful variety and helped improve the classifier.

All 44,112 synthetic images were saved as JPEG files in per-class subdirectories with clean absolute paths to avoid path-resolution issues.

VII. EXPERIMENT C — GAN-AUGMENTED CNN

A. Dataset Composition

The final training set for Experiment C merges all real training images with the per-class synthetic images:

- Real training images: 6,270
- Synthetic images (GAN): 44,112

- Combined training set: **50,382 images**
- Per-class balance: exactly 6,705 images each

With perfectly balanced data, the CrossEntropyLoss class weights become near-uniform (ranging 0.94 to 1.06 across classes, compared to 0.04 to 2.8 in Experiments A and B). This simplifies the optimization landscape considerably.

B. Training Behavior

A notable difference from Experiments A and B is that the validation loss in Phase 1 already starts at a reasonable level (0.977 at epoch 1) because the model is trained on a well-balanced dataset. By the end of Phase 1, training accuracy reached 59% and validation accuracy 66.8%. Phase 2 training is much more stable than in the imbalanced experiments: the training loss decreases steadily and the validation accuracy reaches 80.5% at its peak (Phase 2, epoch 12), compared to 82.1% for A and 68.1% for B. More importantly, the classification metrics on rare classes improve dramatically.

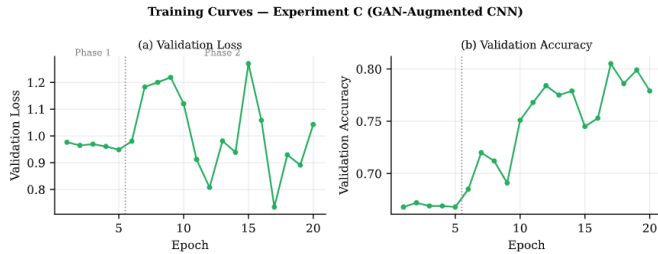


Fig. 6. Validation loss and accuracy for Experiment C. Training is noticeably more stable than Experiments A and B.

C. Results

Experiment C achieves a macro F1 of **0.6680**, a gain of +18.87 percentage points over A and +19.47 over B. Every class improves over both baselines without exception.

VIII. FINAL COMPARISON AND ANALYSIS

A. Full Results Table

Table VI and Figure 7 present the per-class F1 scores and macro F1 for all three experiments.

TABLE VI
FINAL PER-CLASS F1 COMPARISON ACROSS ALL THREE EXPERIMENTS

Class	Full Name	A	B	C
akiec	Actinic Keratosis	0.286	0.338	0.463
bcc	Basal Cell Carc.	0.554	0.687	0.784
bkl	Benign Keratosis	0.503	0.459	0.539
df	Dermatofibroma	0.385	0.140	0.625
mel	Melanoma	0.283	0.372	0.514
nv	Melanocytic Nevi	0.678	0.682	0.905
vasc	Vascular Lesion	0.667	0.636	0.846
Macro Avg		0.479	0.473	0.668

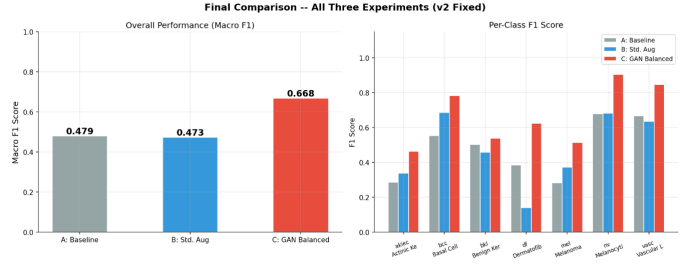


Fig. 7. Final comparison: (a) per-class F1 and (b) macro F1. Experiment C (green) outperforms both baselines on every class.

B. Key Observations

Observation 1 — GAN augmentation wins on every class. Experiment C outperforms both A and B on all seven classes. This is not just a minority-class effect; even the majority class *nv* improves from 0.678 to 0.905. When the model is trained on balanced data, it learns more discriminative features for every class because no single class dominates the gradient updates.

Observation 2 — Standard augmentation has a hard floor. The B vs. A delta column shows that standard augmentation helped some classes (bcc: +0.133, mel: +0.088) but hurt others (df: -0.245, bkl: -0.044). The net macro F1 change was just -0.006 — effectively no improvement over baseline. This confirms the ceiling effect: augmentation transforms cannot substitute for genuinely diverse training data.

Observation 3 — The rarest class benefits most. The *df* class (33 real images, 6,672 synthetic) shows the largest absolute gain: F1 from 0.385 (A) to 0.625 (C), $\Delta = +0.240$. The comparison with Experiment B is even more striking: B actually broke this class (F1 = 0.140), while C fixed it ($\Delta_{C \text{ vs } B} = +0.485$).

Observation 4 — Balanced training stabilizes optimization. In Experiment C, training loss and validation loss track much more closely than in A or B. The extreme overfitting seen in Experiment A (training loss < 0.01, validation loss > 1.5 in Phase 2) disappears when training data is balanced. This makes sense: the loss landscape is simpler when all classes contribute roughly equally to every gradient update.

Observation 5 — Melanoma recall nearly doubles. *Melanoma* F1 improved from 0.283 to 0.514 ($\Delta = +0.231$). Since missed melanoma has direct life-threatening consequences, this is arguably the most clinically significant result of the project.

IX. CONCLUSION AND FUTURE WORK

This project showed that training one DCGAN per class and fully balancing the HAM10000 training data is significantly more effective than standard data augmentation for skin lesion classification. By generating 44,112 synthetic images across 6 classes and bringing all 7 classes to equal size (6,705 images each), we achieved a macro F1 of 0.6680 — nearly 20 percentage points above both the baseline (0.4793) and

standard augmentation (0.4733). Every single class improved, with the rarest class (*df*) gaining +0.240 in F1 over the baseline and recovering from a catastrophic failure under standard augmentation.

The project makes three concrete contributions. First, a systematic end-to-end pipeline for class-conditional image generation and classification across five clean Jupyter notebooks. Second, a rigorous three-way controlled comparison that isolates the effect of the augmentation strategy from all other variables. Third, a documented failure case for standard augmentation (the *df* class with 33 images) that clearly establishes *when* GAN augmentation becomes necessary rather than merely helpful — roughly when class sizes fall below approximately 50 to 100 real images.

Future improvements. The most impactful next step would be replacing DCGAN with a conditional diffusion model [7], which would produce 224×224 synthetic images and eliminate the upsampling artifact introduced when resizing 64×64 GAN outputs. A single class-conditional GAN trained on all classes simultaneously could also allow knowledge sharing between visually similar lesion types, making the generation more parameter-efficient. The Grad-CAM visualizations computed in Experiment C showed that the GAN-balanced model focuses more consistently on the actual lesion region rather than surrounding skin — this interpretability improvement is worth formalizing as a quantitative metric in future work. Finally, applying xLSTM [6] style temporal modeling could help if the project were extended to video dermoscopy, where frame-level dependencies carry diagnostic information beyond any single frame.

ACKNOWLEDGMENT

This project was inspired by research discussed at NeurIPS 2024 and 2025, specifically the Test-of-Time Award for the original GAN paper, the Best Paper Award for Visual Autoregressive Modeling, the xLSTM spotlight presentation, and the NeurIPS 2025 Best Paper on diffusion model generalization. The dataset was provided by the HAM10000 authors [1].

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